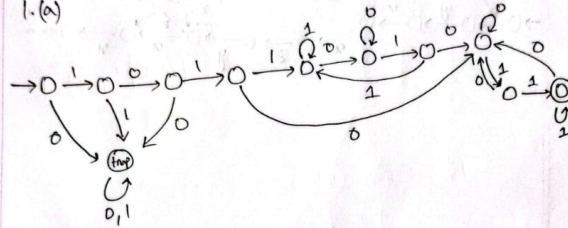


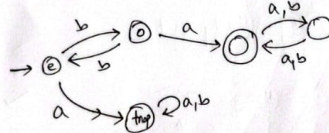
TOC

Solution to Mid, Fall 2025

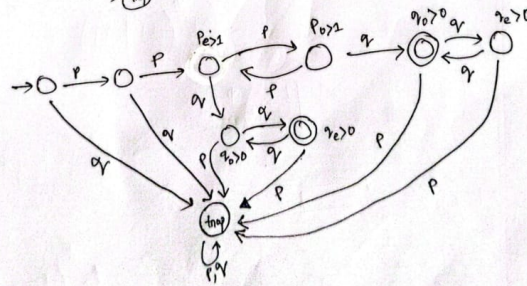
1.(a)



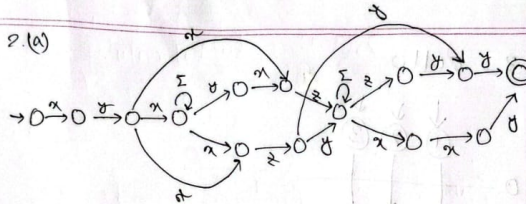
(b)



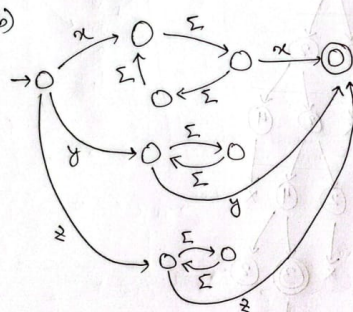
(c)



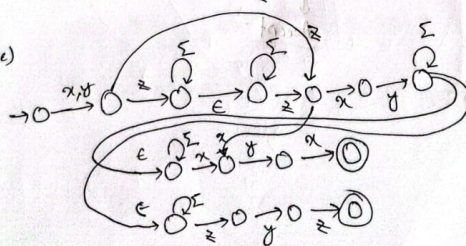
2.(a)



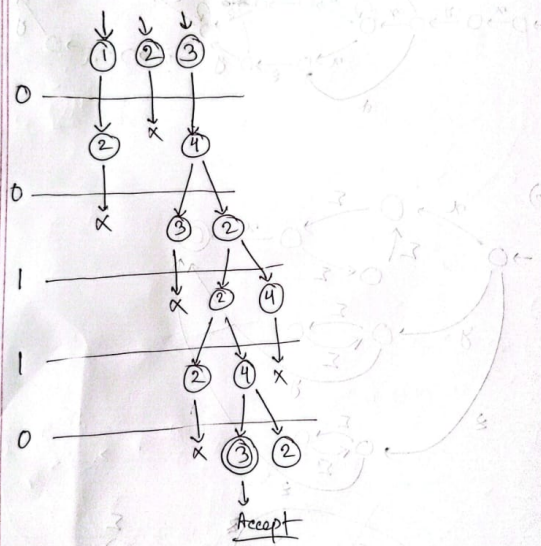
(b)



(c)



3. 00110



4.

	0	1
$\rightarrow \{A\}$	$\{A, B\}$	$\{A, F\}$
$\{A, B\}$	$\{A, B, F\}$	$\{A, C, F\}$
$* \{A, F\}$	$\{A, B, F\}$	$\{A, F\}$
$* \{A, B, F\}$	$\{A, B, F\}$	$\{A, C, F\}$
$* \{A, C, F\}$	$\{A, B, F\}$	$\{A, B, F\}$

5. (a) $1^* 0 1^* 0 (1^* 0)^* 1^*$

(b) $y^* (xy^+ | xxy^+)^* (e | x | xx)$

(c) $pqr \Sigma^* p p q r \Sigma^* q p q r \mid p q \Sigma^* p p q p q r$